

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: KEYMAKER-U01 Umbrella Master Study: Studies of Investigational Agents With Either Pembrolizumab (MK-3475) Alone or With Pembrolizumab PLUS Chemotherapy in Participants With Non-small Cell Lung Cancer (NSCLC) (MK-3475-U01/KEYMAKER-U01) **Short Title/Acronym:** KEYMAKER-U01 (MK-3475-U01) **Protocol Number:** MK-3475-U01 **NCT Number:** NCT04165798 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Pembrolizumab (MK-3475) and various investigational agents (with or without chemotherapy) **Phase of Development:** Phase 2 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (primarily Objective Response Rate [ORR]) of different investigational agents in combination with pembrolizumab given in sequence or in combination with pembrolizumab PLUS chemotherapy in the treatment of advanced NSCLC. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Duration of Response (DOR), Overall Survival (OS), and the safety and tolerability of the different treatment combinations.

2.2 Background & Rationale The anti-PD-1 monoclonal antibody pembrolizumab, with or without chemotherapy, is a standard of care for first-line advanced or metastatic NSCLC. However, there is a critical need to improve response rates and provide options for patients whose disease progresses. This umbrella master protocol uses a platform design consisting of a master screening study and multiple rolling-arm substudies. It efficiently evaluates multiple novel investigational agents in diverse NSCLC populations (both treatment-naïve and previously treated) to rapidly identify promising combination therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Umbrella Master Protocol / Platform Design (rolling arms) **Blinding:** Open-label (varies by substudy) **Randomization:** Randomized (across specific substudies/arms)

3.2 Planned Enrollment & Duration Target **NS\$:** Approximately 1,065 participants **Number of Sites:** ~40 Global Locations **Estimated Study Start:** 19-Dec-2019 **Estimated Primary Completion:** 13-Feb-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically- or cytologically-confirmed diagnosis of Stage IV squamous or nonsquamous NSCLC. 3. Has measurable disease per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1).

4.2 Exclusion Criteria 1. Has an active autoimmune disease that has required systemic treatment in the past 2 years. 2. Has a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 3. Has an active infection requiring systemic therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Varies by assigned substudy arm. Core backbone includes Pembrolizumab 200 mg administered every 3 weeks (Q3W). This is combined with specific investigational agents (e.g., MK-4830, MK-0482, vibostolimab, boserolimab) and/or standard chemotherapy regimens. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 35 cycles (approximately 2 years) or until disease progression/unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant standard-of-care symptom management and chemotherapy as designated by specific substudy arms (e.g., carboplatin, paclitaxel). **Prohibited:** Concurrent treatment with systemic steroids (except for physiological replacement or management of immune-related adverse events), live vaccines, and any other unapproved investigational anticancer agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Tissue/Blood Sample Collection
Baseline	Day 0	Enrollment, Randomization to Substudy, First Dose of Pembrolizumab + Investigational Agent
Interim	Q3W $\pm$ 3 Days	Treatment Administration, Safety Labs, AE Monitoring, PK/PD Sampling
End of Study	Up to Cycle 35	Final Safety Assessments, Exit Interview, Transition to Survival Follow-up

(Note: Tumor imaging occurs at protocol-defined intervals, typically every 9 weeks)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR), defined as the proportion of patients achieving a Complete Response (CR) or Partial Response (PR). **Measurement Method:** Evaluated per RECIST version 1.1 by investigator review and/or blinded independent central review (BICR).

7.2 Secondary & Exploratory Endpoints 1. Progression-Free Survival (PFS) per RECIST 1.1. 2. Duration of Response (DOR) and Time to Response. 3. Overall Survival (OS) and Safety/tolerability profiling (AEs and treatment discontinuations due to AEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during Q3W visits; graded according to NCI CTCAE. Specialized monitoring for immune-related adverse events (irAEs) characteristic of PD-1 inhibitors (e.g., pneumonitis, colitis, endocrinopathies).

Serious Adverse Events (SAEs): Reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An external, independent DMC evaluates ongoing safety and efficacy data across all rolling arms to guide adaptive platform decisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The Intent-to-Treat (ITT) population is used for efficacy analyses. The All-Patients-As-Treated (APaT) population is used for safety summaries. **Sample Size**

Justification: The platform design accommodates flexible enrollment (~1,065 total), powered dynamically per substudy to detect meaningful efficacy signals (ORR) for novel combinations versus historical controls or active comparators. **Handling of**

Missing Data: Participants without post-baseline tumor assessments are generally considered non-responders for ORR. Time-to-event endpoints (PFS, OS) use Kaplan-Meier methodology with predefined censoring rules for missing data or new anti-cancer therapy initiation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits by the sponsor to verify data integrity, protocol adherence, and proper AE reporting. **Audit Trail:** Secure electronic Data Capture

(eDC) system with full 21 CFR Part 11 compliant audit trails, electronic signatures, and data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: MK-3475-U01 **Registry Number:** NCT04165798 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** KEYMAKER-U01 Umbrella Master Study **Official Regulatory Title:** KEYMAKER-U01 Umbrella Master Study: Studies of Investigational Agents With Either Pembrolizumab (MK-3475) Alone or With Pembrolizumab PLUS Chemotherapy in Participants With Non-small Cell Lung Cancer (NSCLC) (MK-3475-U01/KEYMAKER-U01)

12. Clinical Framework (Oncology Specific)

Primary Condition: Stage IV Non-Small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Histologically/cytologically-confirmed advanced NSCLC with measurable disease **Medical Methodology:** Immune Checkpoint Inhibition (PD-1 blockade) combined with novel investigational therapeutics **Optimization Target:** Maximization of anti-tumor immune response and Objective Response Rate (ORR)

13. Study Procedures & Methodology

Management Pathway: Umbrella platform trial structure matching biomarker/clinical profiles to specific substudies **Education Protocol (HCP):** Site initiation and continuous training on managing immune-related AEs (irAEs) and complex multi-agent toxicities **Education Protocol (Patient):** Comprehensive informed consent regarding platform design, potential substudy assignment, and reporting immune-related symptoms **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of imaging scans to ensure unbiased RECIST 1.1 evaluations

14. Observation & Follow-Up Schedule

Total Duration: Up to 35 cycles (~2 years) of treatment + long-term survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Q3W) corresponding to IV infusion cycles **Remote Monitoring Frequency:** Routine safety follow-ups post-treatment discontinuation **Checklist Review Interval:** Tumor imaging assessments generally conducted every 9 weeks for the first year, and every 12 weeks thereafter

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		[Insert Name]		_____		[DD-MMM-YYYY]										
	Clinical Operations Lead		[Insert Name]		_____		[DD-MMM-YYYY]										
		Lead Statistician		[Insert Name]		_____		[DD-MMM-YYYY]									